

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,**  
**LONERE – RAIGAD -402 103**  
**Winter Semester Examination – December - 2019**

**Branch: B. Tech Mechanical**

**Subject with Code: - Theory of Machine -II (BTMEC504)**

**Date:- 16/12/2019**

**Sem.:- V**

**Marks: 60**

**Time:- 3 Hr.**

**Instructions to the Students**

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately Assume it and should mention it clearly

**(Marks)**

**Que.1.**

**(2×6=12)**

A) A pulley is driven by a flat belt, angle of lap being  $120^\circ$ . The belt is 100 mm wide by 6 mm thick and density  $1000 \text{ kg/m}^3$ . If coefficient of friction is 0.3 & maximum stress in belt is not exceed 2 MPa. Find greatest power which the belt can transmit & corresponding speed of belt.

B) Explain centrifugal Tension in belt.

**Que.2.**

**(2×6=12)**

A) What do you understand by the term 'Interference' & 'undercutting' as applied to gears

B) A pinion of 20 involutes teeth and 125mm pitch circle diameter drives a rack. The addendum of both pinion and rack is 6.25mm. What is the least pressure angle which can be used to avoid interference? With this pressure angle, find the length of the arc of contact and the minimum numbers of teeth in contact at a time.

**Que.3.** Attempt the following

A) Explain compound gear train with neat sketch.

**(6)**

B) In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 rpm in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 rpm in the clockwise direction, what will be the speed of gear B.

**(6)**

**Que. 4.**

**(2×6=12)**

A) The turning moment diagram for a petrol engine is drawn to the following scales : Turning moment, 1 mm = 5 N-m Crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the areas above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm<sup>2</sup>. The rotating parts are equivalent to a mass of 36kg at a radius of gyration of 150mm. Determine the coefficient of fluctuation of speed when the engine runs at 1800 r.p.m.

B) Explain the terms. i) Sensitiveness of governor. ii) Stability of governor.  
iii) Isochronism. iv) Hunting of governor.

**Que. 5.**

**(2×6=12)**

A) The mass of the turbine rotor of a ship is 20 tonnes and has a radius of gyration of 0.6m Its speed is 2000 r.p.m. The ship pitches 6° above and 6° below the horizontal position. The complete oscillation takes 30 seconds and the motion is simple harmonic. Determine the following. i) Maximum gyroscopic couple. ii) Maximum angular acceleration of the ship during pitching and iii) The direction in which the bow will tend to turn when rising, if the rotation of the rotor is clockwise when looking from the left.

B) The turbine rotor of ship has a mass of 2000kg and rotates at a speed of 3000r.p.m. clockwise when looking from a stern. The radius of gyration of the rotor is 0.5m. Determine the gyroscopic couple and its effect upon the ship when the ship is steering to the right in curve of 100m radius at a speed of 16.1 knots (1 knot = 1855 m/h) calculate also the torque and its effects when the ship is pitching in simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 50 seconds and the total angular displacement between the two extreme positions of pitching is 12°. Find the maximum acceleration during pitching motion.

**Que. 6. Solve the following.**

A) Explain Rayleigh's method. **(6)**

B) Explain and derive expression for critical or whirling speed of a shaft. **(6)**

**\*\*\*\*\*End Paper\*\*\*\*\***

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Winter Examination – 2022**

**Course: B. Tech.                      Branch : Mechanical Engineering                      Semester : V**

**Subject Code & Name: BTMC503 Theory of Machines- II**

**Max Marks: 60**

**Date:02/02/2023**

**Duration: 3.00 Hrs.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO)      Marks

**Q. 1 Solve any two of the following.**

**12**

**A)** Explain with Figure types of Flat belt drives.

Understand

**4**

Compare flat and V-Belt drive.

/CO1

**2**

**B)** Derive the formula for Ratio of Driving tensions in Flat belt drive.

Evaluate/  
CO1

**6**

**C)** The power is transmitted from pulley 1.5m diameter running at 210 RPM to a pulley 2.25 m diameter by means of belt. Find the speed lost by driven pulley as a result of creep, if the stress on tight and slack side of belt is 1.5Mpa and 0.5 Mpa respectively the young's modulus for belt material is 100Mpa.

Evaluate/  
CO1

**6**

**Q.2 Solve any two of the following.**

**12**

**A)** Explain terminology of helical gear with diagram.

Understand  
/ CO2

**6**

**B)** A pinion having 35 teeth drives gear having 80 teeth. The profile of gears is involute with 20° pressure angle, 12mm module and 10mm addendum Find the length of path of contact, arc of contact and contact ratio.

Evaluate/  
CO2

**6**

**C)** State and prove law of gearing.

Evaluate/  
CO2

**6**

**Q. 3 Solve any two of the following.**

**12**

**A)** Explain types of gear trains with neat sketch.

Understand  
/ CO2

**6**

|             |                                                                                                                                                                                                                                                                                                                                        |                    |           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------|
| <b>B)</b>   | In an Epicyclic gear train an arm carries two gears A and B having 36 to 45 teeth respectively if the arm rotates at 150 rpm in anticlockwise direction about centre of gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed makes 300 rpm in clockwise direction, what will be speed of gear B? | Evaluate/<br>CO2   | <b>6</b>  |
| <b>C)</b>   | Explain differential gear box and draw table of motion for it.                                                                                                                                                                                                                                                                         | Understand<br>/CO2 | <b>6</b>  |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                 |                    | <b>12</b> |
| <b>A)</b>   | Give detailed classification of governors.                                                                                                                                                                                                                                                                                             | Understand<br>/CO3 | <b>6</b>  |
| <b>B)</b>   | Explain coefficient of fluctuation of Energy and speed in case of flywheel.                                                                                                                                                                                                                                                            | Understand<br>/CO4 | <b>6</b>  |
| <b>C)</b>   | Explain the effect of Gyroscopic couple on Naval ship during Steering, Pitching and rolling.                                                                                                                                                                                                                                           | Understand<br>/CO5 | <b>6</b>  |
| <b>Q. 5</b> | <b>Solve any two of the following.</b>                                                                                                                                                                                                                                                                                                 |                    | <b>12</b> |
| <b>A)</b>   | Explain the terms longitudinal vibrations, transverse vibrations and torsional vibrations with neat sketch and application of each.                                                                                                                                                                                                    | Understand<br>/CO6 | <b>6</b>  |
| <b>B)</b>   | Compare natural (free) vibration with forced vibration. Also provide two examples for each.                                                                                                                                                                                                                                            | Apply/<br>CO6      | <b>6</b>  |
| <b>C)</b>   | A cantilever shaft 50 mm diameter and 300 mm long has a disc of mass 100 kg at its free end. The Young's modulus for the shaft material is 200 GN/m. Determine the frequency of longitudinal and transverse vibrations of the shaft.                                                                                                   | Evaluate/<br>CO6   | <b>6</b>  |

**\*\*\* End \*\*\***

Branch: Mechanical Engg. (Third Year B.Tech.)  
Subject with Subject Code:- Theory of Machines -II

Sem.: - V  
(BTMEC504)

Marks: 60

Date:- January 2023

Time:- 3 Hrs.

**Instructions:-**

1. Figures to the right indicate full marks.
2. Clearly mention the main question number along with the sub questions.
3. Assume suitable data, if necessary.
4. All questions are compulsory.

| Q.No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks          | CO                |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|
| 1     | Solve <b>Any Two</b> :<br>a) A belt drive transmits 8 kW of power from a shaft rotating at 240 rpm to another shaft rotating at 160 rpm. The belt is 8 mm thick. The diameter of smaller pulley is 600 mm & the two shafts are 5 m apart. The coefficient of friction is 0.25. If the maximum stress in the belt is limited to 3 N/mm <sup>2</sup> , find the width of the belt for<br>i. An open belt drive, and<br>ii. A crossed belt drive<br>b) What are the different types of pulleys? Explain briefly.<br>c) Derive the relation for ratio of belt tensions in a flat belt drive | 06<br>06<br>06 | CO1<br>CO1<br>CO1 |
| 2     | Solve <b>Any Two</b> :<br>a) State & derive law of gearing with neat sketch.<br>b) Make a comparison of Cycloidal & Involute tooth forms.<br>c) A pair of spur gears consists of a 20 teeth pinion meshing with a 120 teeth gear. The module is 4 mm. Calculate<br>(i) the centre distance;<br>(ii) the pitch circle diameters of the pinion and the gear;<br>(iii) the addendum and dedendum;<br>(iv) the tooth thickness;<br>(v) the gear ratio.                                                                                                                                      | 06<br>06<br>06 | CO2<br>CO2<br>CO2 |
| 3     | Solve <b>Any Two</b> :<br>a) What is a gear train? What are its main types (with sketches)?<br>b) Sketch a automotive differential gear box & explain its working.<br>c) An epicyclic gear train is shown in Fig.1. The number of teeth on A & B are 80 & 200. Determine the speed of the arm <i>a</i> .<br>(i) If A rotates 100 rpm clockwise & B at 50 rpm counter-clockwise.<br>(ii) If A rotates 100 rpm clockwise & B is stationary.                                                                                                                                               | 06<br>06<br>06 | CO2<br>CO2<br>CO2 |

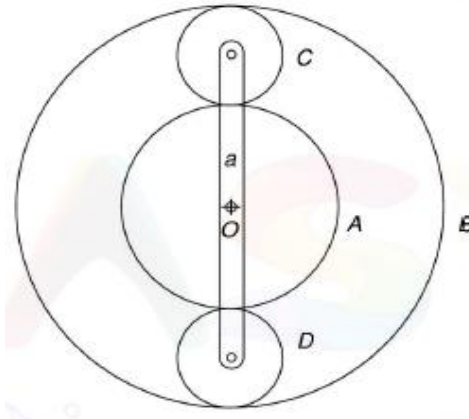


Fig.1

**4 Solve Any Two:**

- a) The turning moment diagram of a multi-cylinder engine is drawn with a scale of  $(1 \text{ mm} = 1^\circ)$  on the abscissa and  $(1 \text{ mm} = 250 \text{ N-m})$  on the ordinate. The intercepted areas between the torque developed by the engine and the mean resisting torque of the machine, taken in order from one end are  $-350, +800, -600, +900, -550, +450$  and  $-650 \text{ mm}^2$ . The engine is running at a mean speed of 750 rpm and the coefficient of speed fluctuations is limited to 0.02. A rimmed flywheel made of grey cast iron FG 200 ( $\rho = 7100 \text{ kg/m}^3$ ) is provided. The spokes, hub and shaft are assumed to contribute 10% of the required moment of inertia. The rim has rectangular cross-section and the ratio of width to thickness is 1.5. Determine the dimensions of the rim.
- b) Explain the terms sensitiveness, hunting, stability, isochronisms & effort of governor relating to the governors.
- c) Sketch a Hartnell governor. Describe its function & deduce a relation to find the stiffness of the spring.

**06 CO4**

**5 Solve Any Two:**

- a) A ship is propelled by turbine rotor having a mass of 6 tonnes & a speed of 2400 rpm. The direction of rotation of the rotor is clockwise when viewed from stern. The radius of gyration of the rotor is 450 mm. Determine the gyroscopic couple & its effect when
- The ship steers to the left in a curve of 60 m radius at a speed of 18 knots ( $1 \text{ knot} = 1860 \text{ m/h}$ )
  - The ship pitches 7.5 degrees above & 7.5 degrees below the normal position & the bow is descending with its maximum velocity; the pitching motion is simple harmonic with a periodic time of 18 seconds.
  - The ship rolls & at the instant, its angular velocity is 0.035 rad/s counter clockwise when viewed from stern
  - Also find the maximum angular acceleration during pitching
- b) Explain in what way the gyroscopic couple affects the motion an aircraft & naval ship?
- c) How do the effects of gyroscopic couple & of centrifugal force make the rider of a two wheeler tilt on one side? Derive a relation for angle of heel.

**06 CO5**

**06 CO5**

**06 CO5**

**6 Solve Any Two:**

- a) Explain the whirling of vertical shaft carrying a single rotor without damping & derive the equation for deflection of the shaft.

**06 CO6**

b) Determine the natural frequency of the system shown in Fig.2.

06 CO6

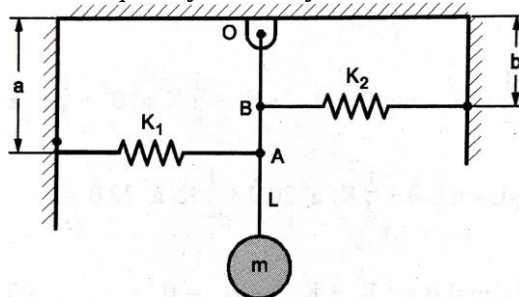


Fig.2.

c) In a forced vibratory system a body having 2 kg mass vibrates in a viscous fluid. The harmonic exciting force of 20 N acting on the mass result in a resonance amplitude of 15 mm with a period of 0.15 sec. Determine the damping coefficient of viscous fluid.

06 CO6

If the system is excited by the same harmonic force but at a frequency of 5 cycles/sec. what will be the amplitudes of forced vibration with & without damper?

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**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Summer Examination – 2023**

**Course: B. Tech.      Branch : Mechanical Engineering      Semester : Fifth**

**Subject Code & Name: BTMC503 Theory of Machines- II**

**Max Marks: 60**

**Date:11/08/2023**

**Duration: 3.00 Hrs.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO)    Marks

**Q. 1 Solve Any Two of the following.**

- |                                                                                                                                                                                                                                                                                                                                                           |                     |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------|
| <b>A)</b> Explain with neat sketch the phenomena of 'slip' and 'creep' in a belt drive.                                                                                                                                                                                                                                                                   | Understand/<br>CO1  | <b>06</b> |
| <b>B)</b> Deduce an expression for the length of belt in an open belt drive.                                                                                                                                                                                                                                                                              | Analyzing /<br>CO1  | <b>06</b> |
| <b>C)</b> A pulley is driven by a flat belt, the angle of lap being $120^\circ$ . The belt is 100 mm wide by 6 mm thick and density $1000 \text{ kg/m}^3$ . If the coefficient of friction is 0.3 and the maximum stress in the belt is not to exceed 2 MPa, find the greatest power which the belt can transmit and the corresponding speed of the belt. | Evaluating /<br>CO1 | <b>06</b> |

**Q.2 Solve Any Two of the following.**

- |                                                                                                                                                                                                                                                   |                     |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------|
| <b>A)</b> Define the terms (i) Module    (ii) Pressure angle    (iii) Addendum<br>(iv) Dedendum    (v) Pitch Circle    (vi) Pitch Circle diameter                                                                                                 | Remember /<br>CO2   | <b>06</b> |
| <b>B)</b> Classify gears on the basis of various shaft positions. Also name the application of each. Provide neat diagram wherever possible.                                                                                                      | Apply / CO2         | <b>06</b> |
| <b>C)</b> A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gears is involute with $20^\circ$ pressure angle, 12 mm module and 10 mm addendum. Find the length of path of contact, arc of contact and the contact ratio. | Evaluating /<br>CO2 | <b>06</b> |

**Q. 3 Solve Any Two of the following.**

- |                                                                                                                                |                     |           |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------|
| <b>A)</b> Explain in short types of gear trains with line diagram. Also state the formula to find velocity ratio in each case. | Understand /<br>CO2 | <b>06</b> |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------|

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |           |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------|
| <b>B)</b>                                   | Describe various types of the torques in an epicyclic gear train and their relationship with neat sketch.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Remember /<br>CO2   | <b>06</b> |
| <b>C)</b>                                   | In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 rpm. in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 rpm. in the clockwise direction, what will be the speed of gear B?                                                                                                                                                                                             | Evaluating /<br>CO2 | <b>06</b> |
| <b>Q.4 Solve Any Two of the following.</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |           |
| <b>A)</b>                                   | Define the following terms relating to governors :<br>(1) Stability                      (2) Sensitiveness                      (3) Isochronism<br>(4) Hunting                      (5) Effort of Governor      (6) Power of Governor                                                                                                                                                                                                                                                                                                                        | Remember /<br>CO3   | <b>06</b> |
| <b>B)</b>                                   | Explain the turning moment diagram of a four stroke cycle internal combustion engine. Also explain the role of flywheel in IC Engine.                                                                                                                                                                                                                                                                                                                                                                                                                        | Understand /<br>CO4 | <b>06</b> |
| <b>C)</b>                                   | The turbine rotor of a ship has a mass of 3500 kg. It has a radius of gyration of 0.45 m and a speed of 3000 rpm clockwise when looking from stern. Determine the gyroscopic couple and its effect upon the ship:<br>1. When the ship is steering to the left on a curve of 100 m radius at a speed of 36 km/h.<br>2. When the ship is pitching in a simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12 degrees. | Applying /<br>CO5   | <b>06</b> |
| <b>Q. 5 Solve Any Two of the following.</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |           |
| <b>A)</b>                                   | Discuss briefly with neat sketch the longitudinal, transverse and torsional free vibration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Remember /<br>CO6   | <b>06</b> |
| <b>B)</b>                                   | Derive an expression for prediction of natural frequency of vibration for single spring mass system.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Understand /<br>CO6 | <b>06</b> |
| <b>C)</b>                                   | Explain and Derive expression for critical or whirling speed of shaft.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Analyzing /<br>CO6  | <b>06</b> |

\*\*\* End \*\*\*

|                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                           |                |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------|
| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Summer Examination-2023</b><br><b>Course: TY B. Tech.                      Branch: Mechanical                      Semester: V</b><br><b>Subject Code &amp; Name: BTMEC504 Theory of Machines II</b><br><b>Max Marks: 60                      Date:17/08/2023                      Duration: 3.00 Hr.</b> |                                                                                                                                                                                                                                                                                                                                                                                           |                |          |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.           |                                                                                                                                                                                                                                                                                                                                                                                           |                |          |
|                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                           | (Level/<br>CO) | Marks    |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |                |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Explain various types of Flat Belt Drives                                                                                                                                                                                                                                                                                                                                                 | <b>CO1/L2</b>  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Define the terms: Slip of Belt, Creep of Belt                                                                                                                                                                                                                                                                                                                                             | <b>CO1/L1</b>  | <b>6</b> |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Derive an expression for length of an Open Belt Drive                                                                                                                                                                                                                                                                                                                                     | <b>CO1/L6</b>  | <b>6</b> |
|                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                           |                |          |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |                |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Classify various types of Toothed Wheels                                                                                                                                                                                                                                                                                                                                                  | <b>CO2/L4</b>  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Explain construction and working of reverted gear train                                                                                                                                                                                                                                                                                                                                   | <b>CO2/L2</b>  | <b>6</b> |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                    | A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gears is involute with 20° pressure angle, 12 mm module and 10 mm addendum. Calculate the length of path of contact, arc of contact and the contact ratio.                                                                                                                                                     | <b>CO2/L3</b>  | <b>6</b> |
|                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                           |                |          |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |                |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Explain construction and working of centrifugal governor                                                                                                                                                                                                                                                                                                                                  | <b>CO3/L2</b>  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Define the terms: Sensitiveness of Governors, Stability of Governors, Isochronous Governors, Hunting of governor, Effort and Power of a Governor                                                                                                                                                                                                                                          | <b>CO3/L1</b>  | <b>6</b> |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                    | A Porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5 kg and the mass of the central load on the sleeve is 25 kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200 mm when the governor is at maximum speed. Calculate the minimum and maximum speeds and range of speed of the governor. | <b>CO3/L3</b>  | <b>6</b> |
|                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                           |                |          |
| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |                |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Derive an expression for Energy Stored in a Flywheel.                                                                                                                                                                                                                                                                                                                                     | <b>CO4/L6</b>  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                    | Explain the concept of gyroscopic couple with neat sketch.                                                                                                                                                                                                                                                                                                                                | <b>CO5/L2</b>  | <b>6</b> |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                    | An aeroplane makes a complete half circle of 50 metres radius, towards left, when flying at 200 km per hr. The rotary engine and the propeller of the plane has a mass of 400 kg and a radius of gyration of 0.3 m. The engine rotates at 2400 r.p.m. clockwise when viewed from the rear. Calculate the gyroscopic couple on the aircraft and state its effect on it.                    | <b>CO5/L3</b>  | <b>6</b> |

|             |                                                                                                                                                                                                                                                                    |               |          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|
|             |                                                                                                                                                                                                                                                                    |               |          |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                             |               |          |
| <b>A)</b>   | Define the terms Used in Vibratory Motion: 1. Period of vibration or time period, 2. Cycle, 3. Frequency                                                                                                                                                           | <b>CO6/L1</b> | <b>6</b> |
| <b>B)</b>   | Explain Critical or Whirling Speed of a Shaft                                                                                                                                                                                                                      | <b>CO6/L2</b> | <b>6</b> |
| <b>C)</b>   | Calculate the whirling speed of a shaft 20 mm diameter and 0.6 m long carrying a mass of 1 kg at its mid-point. The density of the shaft material is 40 Mg/m <sup>3</sup> , and Young's modulus is 200 GN/m <sup>2</sup> . Assume the shaft to be freely supported | <b>CO6/L3</b> | <b>6</b> |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                 |               |          |

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                   | (Level/CO) | Marks     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q.1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                        |            | <b>12</b> |
| A) Explain the following types of pulleys with neat sketch ,<br>i. idler pulleys<br>ii. intermediate pulleys<br>iii. loose & fast pulleys<br>iv. guide pulleys                                                                                                                                                                                                                                    | CO1        | 6         |
| B) What are different modes of transmitting power from one shaft to another? Compare them.                                                                                                                                                                                                                                                                                                        | CO1        | 6         |
| C) A 100 mm & 10 mm thick belt transmits 5 kW of power between two parallel shafts. The distance between shaft centers is 1.5 m & diameter of the smaller pulley is 440 mm. The driving & driven shafts rotate at 60 rpm & 150 rpm respectively. The coefficient of friction is 0.22. Find the stress in the belt if the two pulleys are connected by<br>i. An open belt; and<br>ii. A cross belt | CO1        | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                        |            | <b>12</b> |
| A) Give a detailed classification of Gears.                                                                                                                                                                                                                                                                                                                                                       | CO2        | 6         |
| B) Define Circular pitch, Addendum, dedendum, pitch circle diameter, module & backlash.                                                                                                                                                                                                                                                                                                           | CO2        | 6         |
| C) Describe the features of involute tooth profile in detail.                                                                                                                                                                                                                                                                                                                                     | CO2        | 6         |
| <b>Q.3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                        |            | <b>12</b> |
| A) What is the difference between a simple gear train & compound gear train? Explain with the help of sketches.                                                                                                                                                                                                                                                                                   | CO2        | 6         |

- B) An epicyclic gear train consist of an arm & two gears A & B having 30 & 40 teeth respectively. The arm rotates about the centre of the gear A at a speed of 80 rpm counter clockwise. Determine the speed of the gear B if,
- the gear A is fixed, and
  - the gear A revolves at 240 rpm clockwise instead of being fixed

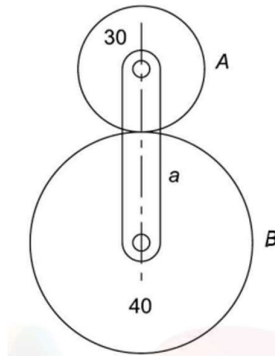


Fig.1

- C) Describe the construction & working of sliding gear box with neat sketch. CO2 6

Q.4 Solve Any Two of the following. 12

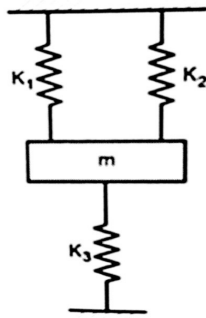
- A) In a Hartnell governor, the extreme radii of rotation of the balls are 40 mm & 60 mm; and the corresponding speeds are 210 rpm & 230 rpm. The mass of each ball is 3 kg. The lengths of the ball & the sleeve arms are equal. Determine the initial compression & the constant of the central spring. CO3 6
- B) Describe the role of flywheel & governor in an automobile. CO4 6
- C) An aero-plane flying at 240 km/h turns towards left & completes a quarter circle of 60 m radius. The mass of rotary engine & the propeller of the plane is 450 kg with a radius of gyration of 320 mm. The engine speed is 2000 rpm clockwise when viewed from the rear. Determine the gyroscopic couple on the aircraft & state its effect. CO5 6

In what way is the effect changed when the

- Aeroplane turns towards right
- Engine rotates clockwise when viewed from the front (nose end) & the aeroplane turns (a) left (b) right?

Q.5 Solve Any Two of the following. 12

- A) What is vibration? What are the various types of vibration? CO6 6
- B) Find the natural frequency of vibration of the system shown in Fig.2. (K<sub>1</sub>= 1000 N/m, K<sub>2</sub> = 1000 N/m, K<sub>3</sub> = 2000 N/m, m =10 kg) CO6 6



**Fig.2.**

**C) Describe types of dampings in detail.**

**CO6**

**6**

**\*\*\* End \*\*\***